Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	If a body A exerts a force on a body B, then B exerts an equal and opposite force on A (1) Accept: For every action there is an <u>equal and opposite</u> reaction	1			1		
		(ii)	1. <u>Gravitational</u> force of drone on Earth (1) (All required) 2. Force of rotors on air (1) (All required) Accept rotors pushing on air		2		2		
	(b)	(i)	The rate of change of momentum of an object is proportional to the resultant force acting on it, (and takes place in the direction of that force) [or equivalent] **Acceptable minimum answers**: Force is proportional (or equal) to the rate of change of momentum / force = <u>change</u> in momentum ÷ time **Or** in symbols e.g. $F = \frac{\Delta p}{\Delta t}$ and Δp defined	1			1		
		(ii)	Substitution into $4\pi r^2 \rho v^2$ i.e. $4 \times \pi \times (5 \times 10^{-2})^2 \times 1.3 \times (22)^2$ (1) F (= weight) = 19.7[7 N] or 19.8 [N] seen (1) Assume answer represents weight of drone, even if not stated. Answer gains both marks	1	1		2	2	

Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
		(iii)	F (for 24 m s^{-1}) = 23.5 [N] **ecf** (1) ΣF = 23.5 **ecf** – 20 (or candidate answer to (b)(ii)) = 3.5 [N] (or 3.7 N) (1) Mass determined = 2.0 k[g] (1) $a = \frac{\Sigma F}{m} = \frac{3.5}{2.0\text{ ecf}}$ = 1.75 m s^{-2} or 1.85 m s^{-2} (1) **unit mark** Accept correct answer based on candidate figures **OR:** Force = 23.5 [N] (obtaining force) (1) Mass = 2.015 k[g] (obtaining mass) (1) $a = \frac{\Sigma F}{m} = \frac{23.5}{2.015}$ = 11.67 and subtracting acceleration due to gravity (1) Answer = 1.85 m s^{-2} (1) **unit mark**	1	1 1 1		4	4	
	(c)	(i)	Rate of change of displacement Accept velocity = $\frac{\text{displacement}}{\text{time}}$ Don't accept distance in a given direction ÷ time **or** displacement in a given period of time	1			1		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Speed (of drone) = $\frac{170}{32}$ = 5.3 [m s^{-1}] (1) Displacement = 78.1 [m] (1) Velocity = $\frac{78.1(\text{ecf})}{32}$ = 2.4 [m s^{-1}] (1) Ted correct (Bill incorrect) - only award mark for correct reasoning and consistent with calculated speed and velocity (1) Accept answers based on distance / displacement only i.e. Distance = 170 [m] (1) Displacement = 78.1 [m] (1) Time same for both stated (or equivalent) (1) Ted correct / Bill incorrect - only award mark if correct reasoning and consistent with calculated distance and displacement (1)			4	4	3	
	(d)		**Benefits (1 from):** • Surveying (general) or filming e.g archaeological sites, volcanoes, disaster areas etc • Monitoring (general) e.g wildlife • Potential for delivery of goods / Amazon • Aerial photography • Able to fly where other aircraft can't **Risks (1 from):** • Delivery of illegal goods e.g prison • Invasion of privacy • Air to air collisions • Disruption to society • Falling out of control • Risk of injury to humans and wildlife [due to the blades]			2	2		
			Question 3 total	**5**	**6**	**6**	**17**	**9**	**0**